

AI-Powered Real Estate Intelligence, Valuation & Investment Recommendation Platform

Forecasting Report

Type	Regression + Recommendation System — Real Estate Intelligence Platform
Dataset	Indian Real Estate Housing Dataset (Kaggle) + Census Population Density Data
Dataset URL	kaggle.com/datasets/ankushpanday1/india-house-price-prediction
Records	2,50,000 Property Records
Features	34 Features (23 Original + 11 Engineered)
Best Model	XGBoost Regressor
Test Accuracy	R ² Score = 0.9769
MAE RMSE	13.68 Lakhs 18.03 Lakhs
Deployment	Streamlit web application
Engine	Forecaster.py

This Forecasting Report presents the future property price projections generated by the platform using historical market trends, location-based growth rates, and investment indicators. The forecasting module estimates property values over 1-year, 3-year, and 5-year horizons to help investors evaluate long-term appreciation potential and make informed decisions. These forecasts provide valuable insights into future market opportunities and support strategic real estate investment planning.

1. Introduction

Real estate investments are inherently long-term holds. While the AI predicts the current fair market value, investors need visibility into future equity gains. The Phase 10 Forecasting Module projects the property's financial appreciation across 1-year, 3-year, and 5-year horizons using historical macroeconomic data.

2. Forecasting Methodology

The methodology relies on Compound Annual Growth Rate (CAGR) assumptions derived from historical Indian city-tier real estate performance. Instead of a blanket national average, the `forecaster.py` engine uses a localized dictionary to assign specific growth rates to tier-1 and tier-2 cities.

3. Growth Rate Model

City Name	Base Annual Growth Rate (CAGR)
Gurgaon	8.5%
Mumbai / Noida	8.0%
Bangalore / Hyderabad	7.5%
Pune	7.0%
New Delhi	6.5%
Chennai	6.0%
Unknown / Default	6.0%

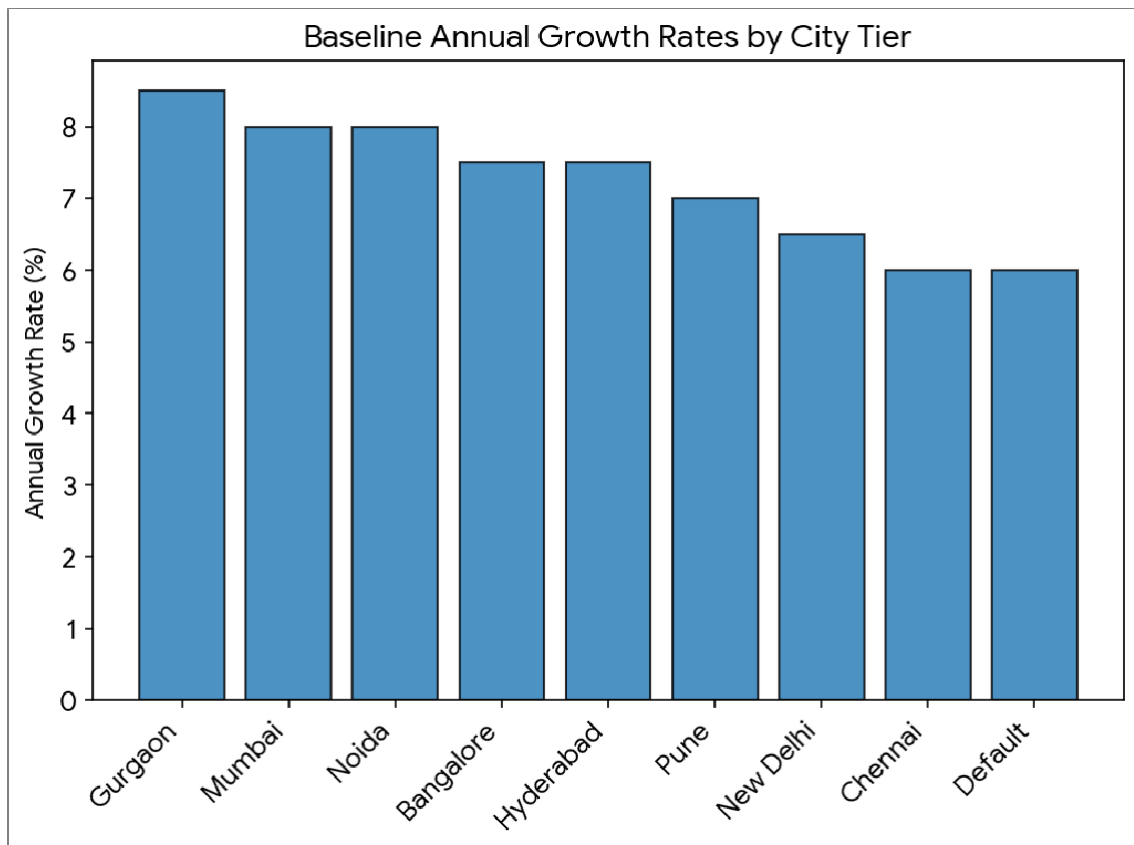


Figure 1: Baseline historical annual growth rates programmed into the forecaster.

Availability Modifier: If a property is marked as `Under_Construction`, the engine adds an execution premium of **+1.5%** to the base rate, reflecting the sharp price correction that occurs upon project completion and handover.

4. Forecast Formula

The core algorithm utilizes standard compounding interest mathematics applied to the AI's predicted current baseline.

Formula: $\text{Future_Price} = \text{Current_Predicted_Price} * (1 + \text{Adjusted_Growth_Rate})^{\text{Years}}$

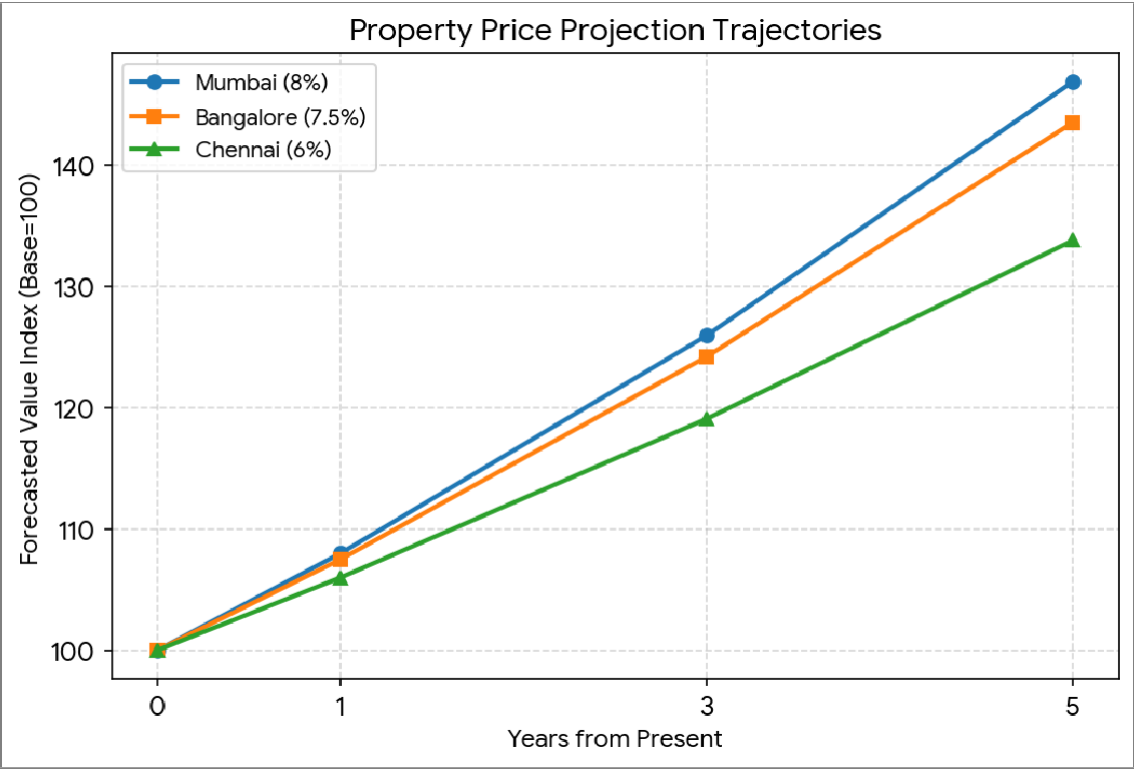


Figure 2: Compounding forecast trajectories for properties across three distinct city tiers.

5. Sample Forecasts

Assuming an AI-predicted baseline price of ₹100 Lakhs for three properties:

City & Status	Adjusted Rate	1-Year Value	3-Year Value	5-Year Value
Mumbai (Ready)	8.0%	₹108.00L	₹125.97L	₹146.93L
Bangalore (Under Const.)	9.0%	₹109.00L	₹129.50L	₹153.86L
Chennai (Ready)	6.0%	₹106.00L	₹119.10L	₹133.82L

6. Assumptions and Limitations

The forecasting model operates under "Ceteris Paribus" (all other things being equal) assumptions:

- Assumes no major macroeconomic shocks (e.g., recessions, pandemic).
- Assumes constant inflation and stable RBI repo interest rates.
- Does not account for hyper-local gentrification spikes that outpace city averages.

7. Dashboard Integration

In `06_forecast.py`, the Streamlit interface visualizes the future valuation data via an interactive `plotly.express.line` chart. The UI clearly displays the delta (total gain in Lakhs) and the percentage appreciation, providing an instant summary of long-term wealth generation.

8. Conclusion

The Forecasting Module seamlessly extends the predictive power of the core machine learning system into the future. By anchoring future projections to the AI's hyper-accurate current valuation baseline ($R^2=0.97$), the platform delivers a comprehensive, end-to-end investment prospectus.